

$$\check{\mathbf{Z}}_{t,k} = \left(\mathbf{I}_{D_Z} - \mathbf{W}_{t,k}^Z \right) \hat{\mathbf{Z}}_t + \mathbf{W}_{t,k}^Z \mathbf{Y}_{t,k}^Z,$$

$$\mathcal{C}_{t,k}^Z = \Phi_Z \left(\check{\mathbf{Z}}_{t,k}, \mathcal{H}_t \right)$$